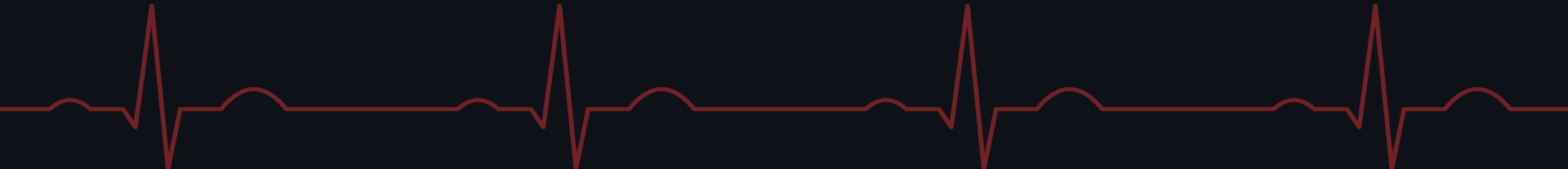


NATURE

*enVast-assisted mechanical thrombectomy versus standard primary PCI
in STEMI with large thrombus burden*



154

PATIENTS

11

CENTRES

1:1

RANDOMISED

-26%

INFARCT SIZE

Large thrombus: the clot that won't stay put

1

Large thrombus burden

Common in STEMI; ballooning and stenting can fragment the clot.

2

Distal embolisation

Debris showers into the microcirculation downstream of the culprit lesion.

3

Microvascular obstruction

Epicardial flow is restored, but tissue-level reperfusion fails.

4

Larger infarct

More myocardium lost, higher risk of heart failure and death.

Aspiration thrombectomy already failed



TASTE

n = 7,244 · no mortality benefit from routine aspiration



TOTAL

n = 10,732 · no clinical benefit; excess stroke at 30 days



Guidelines

Routine thrombus aspiration not recommended (ESC Class III)

enVast: a stent-retriever built for coronary clot

Developed by Vesalio from its NeVa neurovascular thrombectomy platform, enVast captures and removes thrombus mechanically before the operator balloons or stents the culprit lesion.



STEP 1

Cross & deploy

Wire the infarct-related artery and deploy the retriever across the thrombus.



STEP 2

Capture

The clot is engaged and held within the device's struts.



STEP 3

Retrieve, then PCI

Withdraw device and clot together, then complete conventional PCI.

Prospective, multicentre, randomised 1 : 1



154 STEMI patients

large thrombus burden · 11 centres



RANDOMISED 1 : 1



enVast thrombectomy
then conventional PCI



Standard of care
conventional primary PCI

Follow-up: CK-MB AUC · CMR at day 3 · 30-day clinical events

HYPOTHESIS

Superiority of enVast on infarct size

PRINCIPAL INVESTIGATOR

Prof. Marco Valgimigli, Cardiocentro
Ticino, Lugano

SPONSOR

Vesalio

REGISTRY

NCT04969471



Enriched for clot; judged on infarct size

KEY INCLUSION

- STEMI undergoing primary PCI
- TIMI thrombus grade ≥ 3 in the infarct-related artery
- If grade 5 (TIMI 0 flow), grade ≥ 3 re-confirmed after wiring

Why enrich? Benefit is most plausible where there is the most clot to embolise.



PRIMARY EFFICACY

Infarct size by CK-MB area under the curve



KEY SECONDARY

Infarct size on cardiac MRI (% LV mass) at day 3



SAFETY

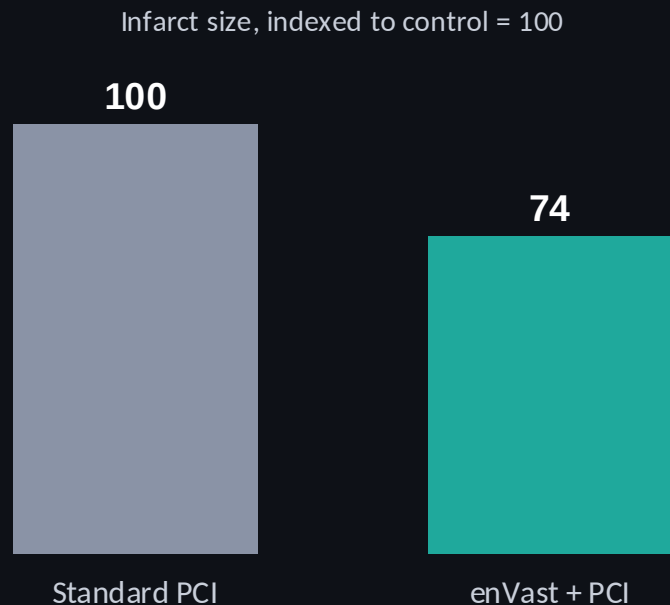
Death, stroke and MACE at 30 days

Infarct size cut by about a quarter

-26%

relative reduction in infarct size (CK-MB AUC) with
enVast versus standard PCI

P = 0.001



Biomarker and MRI tell the same story

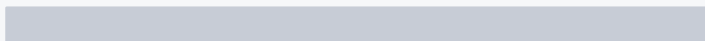


CK-MB AUC

Primary endpoint

-26%

Standard PCI



enVast + PCI



P = 0.001 • significant

≈

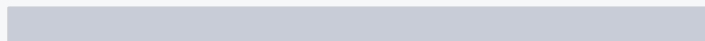


Cardiac MRI, day 3

Infarct size, % LV mass

-25%

Standard PCI

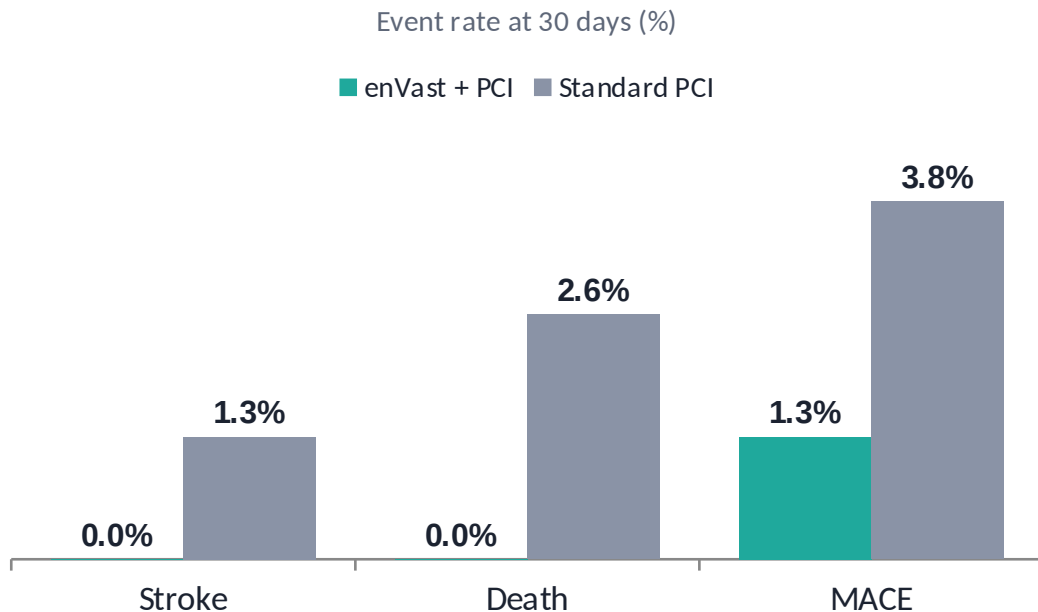


enVast + PCI



Directionally consistent trend

No strokes, no deaths in the enVast arm



Why it matters

Aspiration in TOTAL carried a stroke signal. NATURE saw none with enVast.



Read with care

Only a handful of events; the trial was not powered for clinical outcomes.

Strong signal, early evidence



Strengths

- Randomised, multicentre design
- Population enriched for large thrombus — where benefit is most plausible
- Biomarker and CMR results agree
- No stroke or death signal at 30 days



Limitations

- Modest sample (n = 154)
- Surrogate primary endpoint, not hard outcomes
- Operators cannot be blinded to device use
- CMR effect a trend; industry-sponsored

VERDICT

Confirms the hypothesis on infarct size; generates the hypothesis for clinical outcomes.

What NATURE means for the cath lab

01

Superior on infarct size

enVast before PCI cut enzymatic infarct size by ~26% in large-thrombus STEMI ($P = 0.001$).

02

Consistent and safe

CMR pointed the same way; no strokes or deaths with enVast at 30 days.

03

Not yet practice-changing

A larger outcomes trial is needed before routine use — selective, not universal.

References



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